

INA181 Current Sense Amplifier Module

Key Features

- High-side or low-side bidirectional current measurement
- Fully analog for fast response
- Differential or single-ended output with common-mode voltage at $\frac{1}{2} V_{DD}$
- Maximum high-side voltage 26V

This module is based on the INA181A1 bi-directional current sense amplifier IC from Texas Instruments. The input current is measured and amplified as an output voltage, which is centered around $\frac{1}{2} V_{DD}$ to be easily sampled by an ADC. The module currently comes in two versions: with 200m Ω shunt resistor and without shunt resistor. The former provides a measurement range of $\pm 400\text{mA}$ at a supply voltage of 3.3V. The measurement range of the latter can be configured by the user.

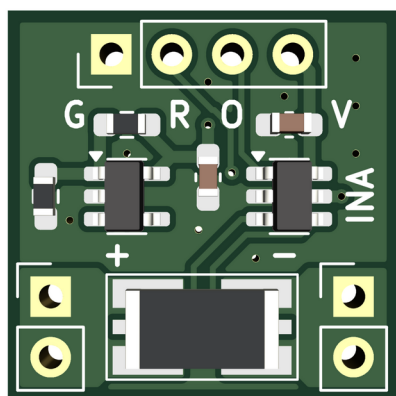


Figure 1: INA181 current sense amplifier module PCB render.

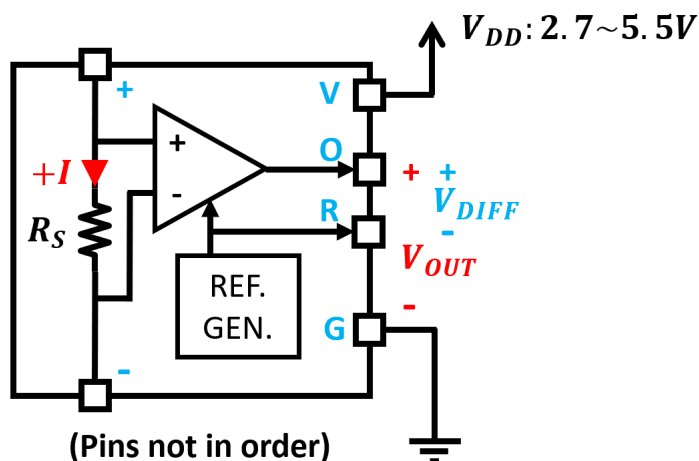


Figure 2: Block diagram of INA181 current sense amplifier module.

Pin Configuration

This module has 6 I/Os:

Pin name	I/O	Description
+	input	Positive terminal of the current measurement.
-	input	Negative terminal of the current measurement.
G	input	Ground connection.
R	output	Reference voltage V_{REF} , $\frac{1}{2} V_{DD}$ is generated by on-board circuitry.
O	output	Current sense amplifier output voltage V_{OUT} .
V	input	V_{DD} supply voltage.

Recommended Operating Condition

		Min.	Max.
V_{DD}	Current sense amplifier supply voltage	2.7V	5.5V
V_{CM}	Input common-mode voltage	-0.2V	26V
I	Measured current (with 200m Ω shunt resistor)	-0.4A	+0.4A

Detailed Description

The module is fully analog, which provides faster response compared to solutions with digital interface such as the INA219 modules. It is recommended to pair the INA181 current sense amplifier module with an MCU and sample the output voltage with the ADC running in continuous mode. This setup best captures the dynamic current.

The on-board reference generator will generate a reference voltage V_{REF} at $\frac{1}{2} V_{DD}$. For example, if the module is supplied by a 3.3V supply, the generated reference voltage V_{REF} will be 1.65V. The output voltage will be centered around the reference voltage V_{REF} . In this example, a single-ended output voltage V_{OUT} of 1.65V will indicate that measured current is 0mA.

The INA181A1 IC offers a voltage gain of 20 V/V. Combined with a current shunt resistor of resistance R_S , the transimpedance gain A_Z of the current sense amplifier is

$$A_Z = 20R_S. \quad (\text{Equation 1})$$

For example, a 200m Ω shunt resistor will offer a combined 4.0 Ω transimpedance gain. The amplified voltage will be centered around the reference voltage V_{REF} .

$$V_{OUT} = V_{REF} + 20IR = \frac{1}{2}V_{DD} + 20IR \quad (\text{Equation 2})$$

With a 200m Ω shunt resistor and supply voltage V_{DD} of 3.3V, the numerical equation for the single-ended output voltage V_{OUT} will be

$$V_{OUT} = 1.65 + 4I. \quad (\text{Equation 3})$$

The following table shows the measurement range and transimpedance gain with different shunt resistor and supply voltage configurations.

Shunt Resistance R_S	Transimpedance Gain A_Z	Measurement Range ($V_{DD} = 3.3V$)	Measurement Range ($V_{DD} = 5V$)
20m Ω	0.4 Ω	$\pm 4.0A$	$\pm 6.25A$
33m Ω	0.66 Ω	$\pm 2.5A$	$\pm 3.8A$
68m Ω	1.36 Ω	$\pm 1.2A$	$\pm 1.8A$
100m Ω	2.0 Ω	$\pm 0.8A$	$\pm 1.25A$
200m Ω	4.0 Ω	$\pm 0.4A$	$\pm 0.625A$

Testing Data

1. DC transfer function ($V_{DD} = 3.3V$). The input is biased at three different common-mode voltages during testing. V_{OUT} is the single-ended output voltage referenced to ground. V_{DIFF} is the differential output voltage referenced to V_{REF} .

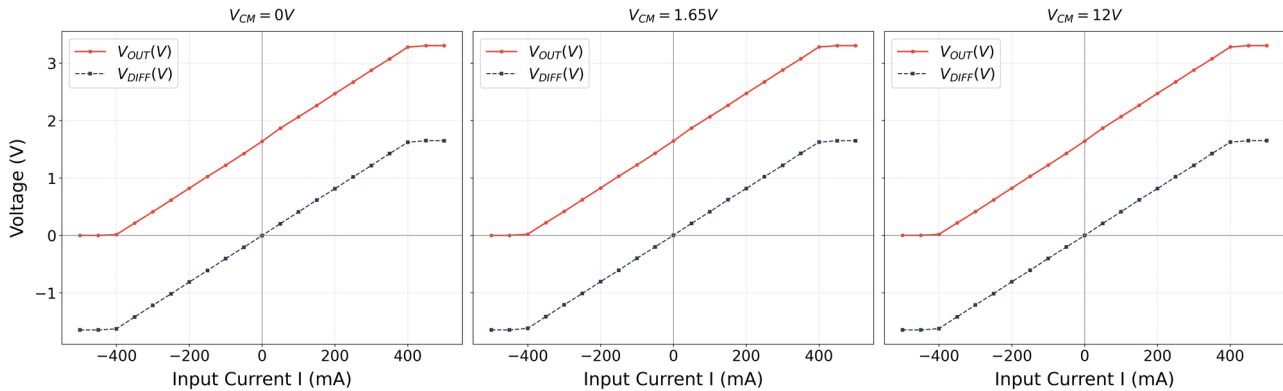


Figure 3: DC transfer function of the INA181 current sense amplifier module.

2. Bi-directional switching current measurement. In this test, the input is subjected to a bi-directionally switching current and the single-ended output voltage V_{OUT} of the INA181 current sense amplifier module is measured. The module is powered by a 3.3V supply and the shunt resistor is 200m Ω . The blue curve shows the output of the module and the orange curve shows the input current waveform.

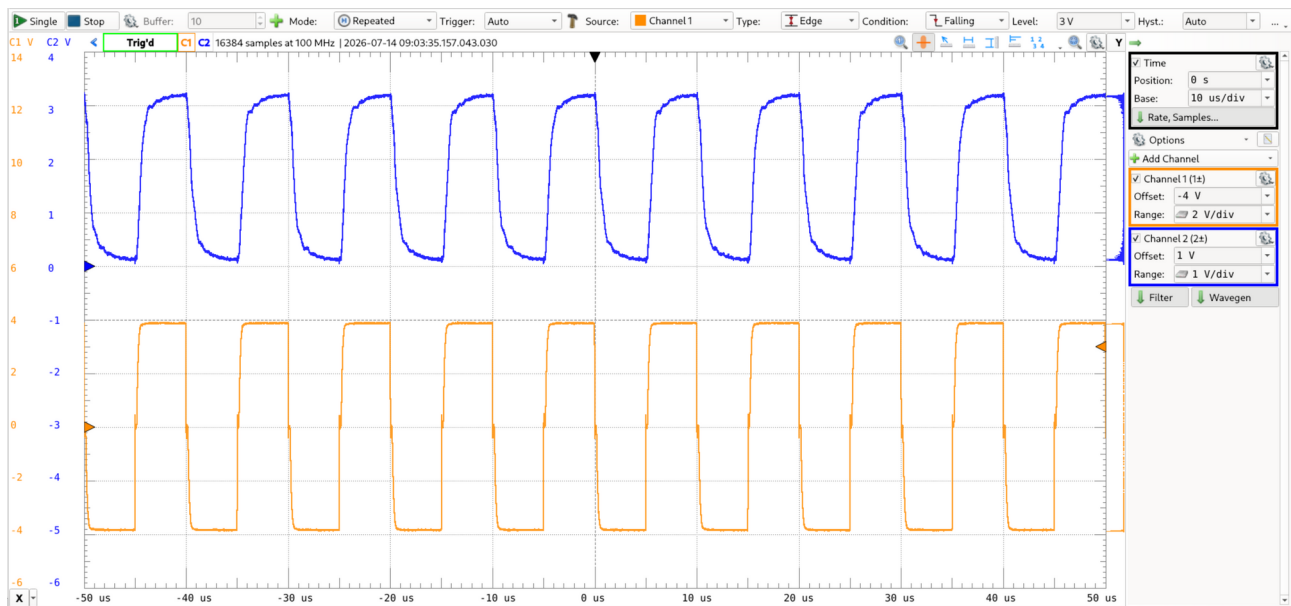


Figure 4: Bi-directionally switching current at 100kHz.

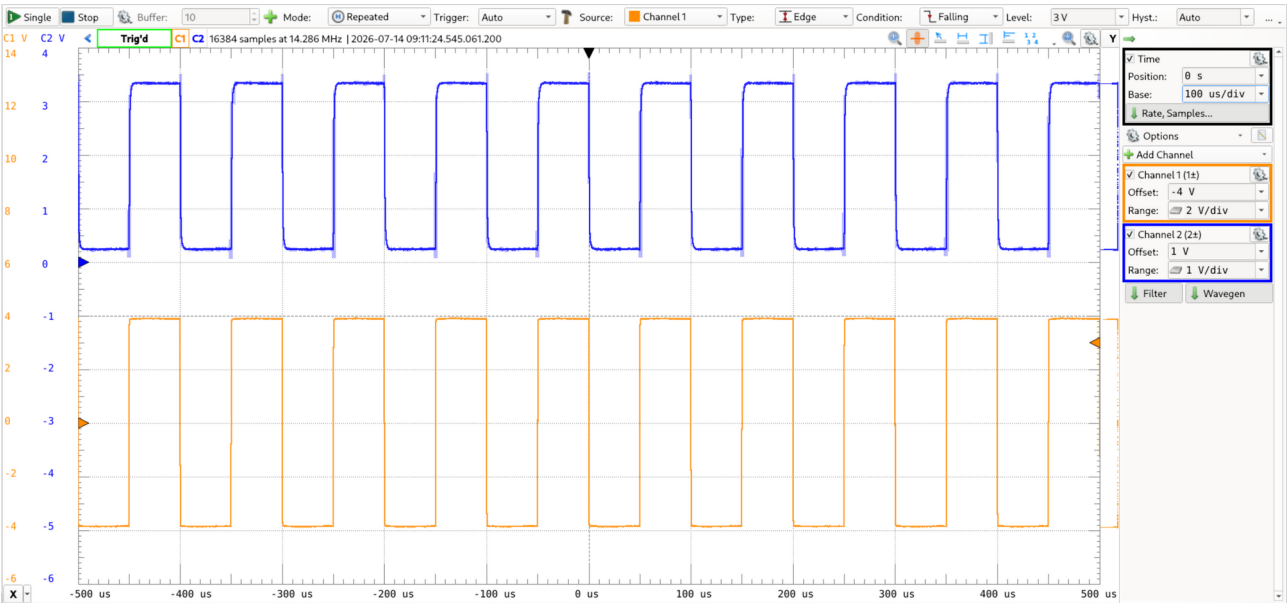


Figure 5: Bi-directionally switching current at 10kHz.

List of Parts

Parts	Bare PCB	With shunt resistor
INA181 Current Sense Amplifier PCB	Included	Included
Shunt Resistor	Not included	200mΩ, soldered
Pin Headers	Not included	Not included

Please contact us if you wish to order the Bare PCB only.

Revision History

PCB Revision	Datasheet Revision	Description
1.0	1.0.0	Initial version.